

# Garv Arora

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## EDUCATION

### Vellore Institute of Technology | Vellore, India

CGPA: 8.37

Bachelor of Technology in Computer Science and Engineering (Specialization: IoT)

Aug. 2023 – July 2027

### Gems International School | Gurugram, India

Class 10: 89.3% | Class 12: 76.4%

Central Board of Secondary Education(CBSE)

Nov. 2016 – April 2023

## TECHNICAL SKILLS

**Development:** Python, C/C++, HTML/CSS, JavaScript, Bash, SQL, MongoDB, Verilog, Java, React, Node.js

**Robotics:** ROS2, Nav2, MoveIt2, RTAB-Map, Kalman Filtering

**Libraries:** OpenCV, MediaPipe, Intel Realsense, PyTorch, TensorFlow, scikit-learn, Flask

**DevOps:** Linux, Docker, Git/GitHub, GitLab CI/CD, AWS

## WORK EXPERIENCE

### Team Vyadh (Martian Rover Team)

April 2024 – Present

Autonomous Systems Developer

Vellore, India

- Developed an autonomous navigation stack using ROS2, Nav2, Visual SLAM using RTAB-Map and Intel RealSense to build elevation maps, enabling real-time path planning and obstacle avoidance.
- Built multithreaded computer vision pipeline with CUDA acceleration, YOLOv8 object detection, ArUco/AprilTag marker recognition and achieved a 4x performance increase using OpenCV and Linux V4L2 optimizations.
- Prototyped Micro-ROS based solutions on ESP32 for sensor acquisition and actuator control and Configured PID control systems for precise robotic mobility with Extended Kalman Filter sensor fusion
- Designed a mission control dashboard using PyQt, providing real-time data visualization and parameter tuning and Integrated Finite State Machine controllers for mission planning

### Wissen Baum Engineering Solutions

May. 2025 – July 2025

Software Automation Intern

Pune, India

- Built BDD automation framework with Python, Behave, and pytest
- Designed parallelized GitLab CI/CD pipelines integrating Playwright and Cypress
- Automated data validation using pandas/NumPy and implemented Git hooks for code quality enforcement

## PROJECTS

### HayaiOS | C, ARM Assembly

2026

- Bare-metal RTOS from scratch implementing preemptive scheduler, context switching, and synchronization
- Featuring Interrupt handling, timer management, and hardware abstraction layer with direct register manipulation

### 3D Reconstruction with IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Implemented real-time 3D surface reconstruction using KinectFusion algorithm with Intel RealSense D455
- Developed IMU-enhanced tracking system using complementary filter fusion to improve camera pose estimation robustness during rapid motion and prevent tracking failures

### PDF Summarizer | Python, PyMuPDF, HuggingFace Transformers, Docker

2025

- Developed persona-based PDF analysis system with RAG-style retrieval, intelligent section ranking, and multi-document processing
- Dockerized automated pipeline using PyMuPDF parsing and Transformer models for context-aware summarization

### Catters – Endless Runner Game | Unity, C#

2024

- Developed Android endless runner with touch-based controls and procedurally generated jungle environments
- Designed complete gameplay systems including scoring, collision detection, and progressive difficulty scaling

## CERTIFICATIONS

**Oracle Cloud Infrastructure 2025:** Generative AI Professional | Data Science Professional

## ACHIEVEMENTS

**IRC 2025:** Ranked 12th globally | **IRDC 2025:** Top 5 Finalist